COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 6 : SLO – 1 | Memory Locations and Addresses (1's & 2's Complement)
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Section A – Match the Following (Answers)

• 1. 1's Complement → B. Flip all bits (0→1, 1→0)
• 2. 2's Complement → A. Add 1 to the 1's complement
• 3. Signed Number Representation → C. Represents negative numbers in binary
• 4. Most Significant Bit (MSB) → D. Indicates sign in signed numbers
• 5. Overflow → E. Error when result exceeds representable range

Section B – Fill in the Blanks

• 1's complement of a binary number is obtained by FLIPPING ALL BITS.
• 2's complement of a number is obtained by taking the 1's complement and then ADDING 1.
• In 2's complement representation, a negative number always begins with 1 in the MSB.
• The range of an 8-bit 2's complement number is from –128 to +127.
• When the result exceeds the possible range of values, it causes an OVERFLOW.

Section C – Activity Task

• Task 1 – 1's and 2's complements:
• a) 0110 1001 → 1's: 1001 0110, 2's: 1001 0111.
• b) 1011 0000 → 1's: 0100 1111, 2's: 0101 0000.
• c) 0001 1110 → 1's: 1110 0001, 2's: 1110 0010.
• Task 2 – Convert to decimal (2's complement method):
• 1111 0100 → negative; 1's = 0000 1011, 2's = 0000 1100 = 12 → value = –12.
• 1000 1111 → negative; 1's = 0111 0000, 2's = 0111 0001 = 113 → value = –113.

Section D – True or False

• 1's complement is obtained by adding 1 to the binary number. — FALSE
• 2's complement is the most widely used method for representing signed numbers. — TRUE
• In 2's complement, there are two representations for zero. — FALSE (only one)
• MSB determines the sign of a signed number. — TRUE
• Taking 2's complement twice returns the original number. — TRUE